

Addendum to Section 3.2: Baseline Hydrology and HRU Areas

Input Page Step 1A Assisted – Baseline Hydrology Module

The user guide provides instructions on how to enter baseline hydrology and HRU area information. The following section describes updates to this section based on the inclusion of a green roof BMP to the WMOST stormwater module capabilities.

The **Land Use** page now shows baseline area, percent effective impervious area (EIA), infiltration rate, roof area, and roof percent EIA under baseline HRU characteristics (see screenshot below). Baseline area, percent EIA, and infiltration rate will still be automatically populated based on the selected watershed model. If you plan to model green roofs, you will need to manually fill in the roof area and roof percent effective impervious. If you do not plan to model green roofs, you can fill the cells with “-9”.

Baseline HRU Characteristics						
HRU ID	HRU Name1	Baseline Area [acre]	Percent Effective Impervious ¹	Infiltration Rate [in/hr] ¹	Roof Area [acre]	Roof Percent Effective Impervious
HRU1B	Forest, Sand and Gravel	3,529	0.0%	9.5	0	0
HRU2B	Open nonresidential, Sand and Gravel	787	4.3%	8.455	10	90
HRU3B	Medium to low density residential, Sand and Gravel	2,567	4.4%	8.075	15	90
HRU4B	High-density residential, Sand and Gravel	361	15.3%	5.95	5	90
HRU5B	Commercial-industrial-transportation, Sand and Gravel	535	65.7%	3.08	10	90

Input Page Step 1B – Land Use and Its Management

After navigating to the **Land Use** page, if you plan to model green roofs through the Stormwater Module, you must also populate HRU roof area and roof percent EIA under baseline HRU characteristics (see screenshot area). If you do not plan to model green roofs, you can fill the cells with “-9”.

Addendum to Section 3.5: Water Supply Sources and Infrastructure

Input Page Step 4A – Surface Water & System Targets

The user guide provides instructions on entering reservoir or surface storage properties and costs. The following section describes how surface water storage is modeled within WMOST, including how to set up a model should your study area not include a surface water storage component.

As noted in the user guide, reservoirs in WMOST may represent reservoirs, lakes or ponds used for water supply, and/or surface storage tanks.¹ The reservoir component within WMOST is conceptually located at

¹The WMOST reservoir component should only be used to model surface water storage tanks if the tanks hold water generated from within the study area. For example, interbasin transfer volumes should not be simulated using WMOST’s reservoir component.

the outlet of the modeled watershed and will receive all of the surface water flows as well as any specified withdrawals and discharges. This conceptual location of the reservoir at the outlet of your study area is relevant for model formulation and calibration and management target specification.

Specifying or Omitting Reservoirs in Model Formulation

With respect to model formulation, if you are modeling a reservoir, constraints must be set for the initial, minimum, and maximum reservoir volumes (see screenshot below).

Initial reservoir/surface storage volume	80	[MG]
Minimum target reservoir/storage volume	33	[MG]
Existing maximum reservoir/storage volume	1,240	[MG]
Initial construction cost	0	[\$/MG]
O&M costs	0	[\$/MG]
Maximum additional reservoir/storage volume	0	[MG]

The definitions for initial, minimum, and maximum reservoir or surface storage volumes can be found on page 41 of the WMOST user guide. If you do not have specific information on the existing maximum reservoir/storage volume, make sure this volume is at least one to two order of magnitudes larger than the initial volume or expected surface water flow to ensure that the reservoir can accommodate the daily flows from the surface water into the reservoir.

If you do not have a reservoir within your study area, enter “0” for the initial, minimum, and maximum reservoir or surface storage volumes to zero out the reservoir, which effectively removes it from the model. If you choose to zero out the reservoir, make sure that your model also excludes the option to purchase virtual water that flows into the reservoir system by setting the associated make-up water penalty to “-9” (see screenshot below). Similarly, you should allow reservoir/sw outflow to be a decision as opposed to a specified data time series (see screenshot below).

Cost for make-up surface water penalty	0	\$/MG
To exclude make-up water penalty, enter -9.		-9
Enter Yes to use Reservoir/Sw Outflow as data time series or No to allow Reservoir/Sw Outflow to be a decision.		No

Accounting for Reservoirs in Model Calibration

With respect to WMOST model calibration, the calibration module compares observed flows and concentrations to modeled flows and concentrations in surface water upstream from the reservoir and its associated components (QSwRes and LSwRes).² Those modeled flows and loadings do not take into account any reservoir withdrawals or discharges (either specified by the user or chosen through optimization). As a result, if you are modeling reservoir withdrawals and discharges, and have measured data that reflects these flows, consider calibrating your model outside of the Calibration Module using flows and loadings leaving the reservoir (DQSwExt and LSwExt).

² WMOST theoretical documentation includes variable definitions and component mass balance equations for reference.

Accounting for Reservoirs in Setting Management Targets

WMOST allows you to set water quality targets for the surface water system (in-stream and reservoir, see WMOST User Guide p. 43). The in-stream concentration and loadings target apply to loadings and concentrations in-stream before the reservoir (LSw, XSwF). The reservoir concentration and loadings target apply to loadings and concentrations within the reservoir (LRes, XResF), which may be different from loadings³ leaving the reservoir at the outlet of the watershed (LSwExt).

Input Page Step 4C – Interbasin Transfer (IBT)

The user guide provides instructions on setting costs and limits on IBTs of water and wastewater on the Interbasin Transfer page. The following section provides additional clarification on the use of this input page.

Overall, the input page is split up into two table sets:

- Table set 1 related to the costs of transferring water in or out of the watershed

If you do not want IBT as a management option, enter -9 for costs AND 0 for flow limits.	
Purchase cost for potable water	670 [\$ /MG]
Purchase cost for wastewater	946 [\$ /MG]
Initial cost for new/increased IBT potable water limit	-9 [\$ /MGD]
Initial cost for new/increased IBT wastewater limit	-9 [\$ /MGD]

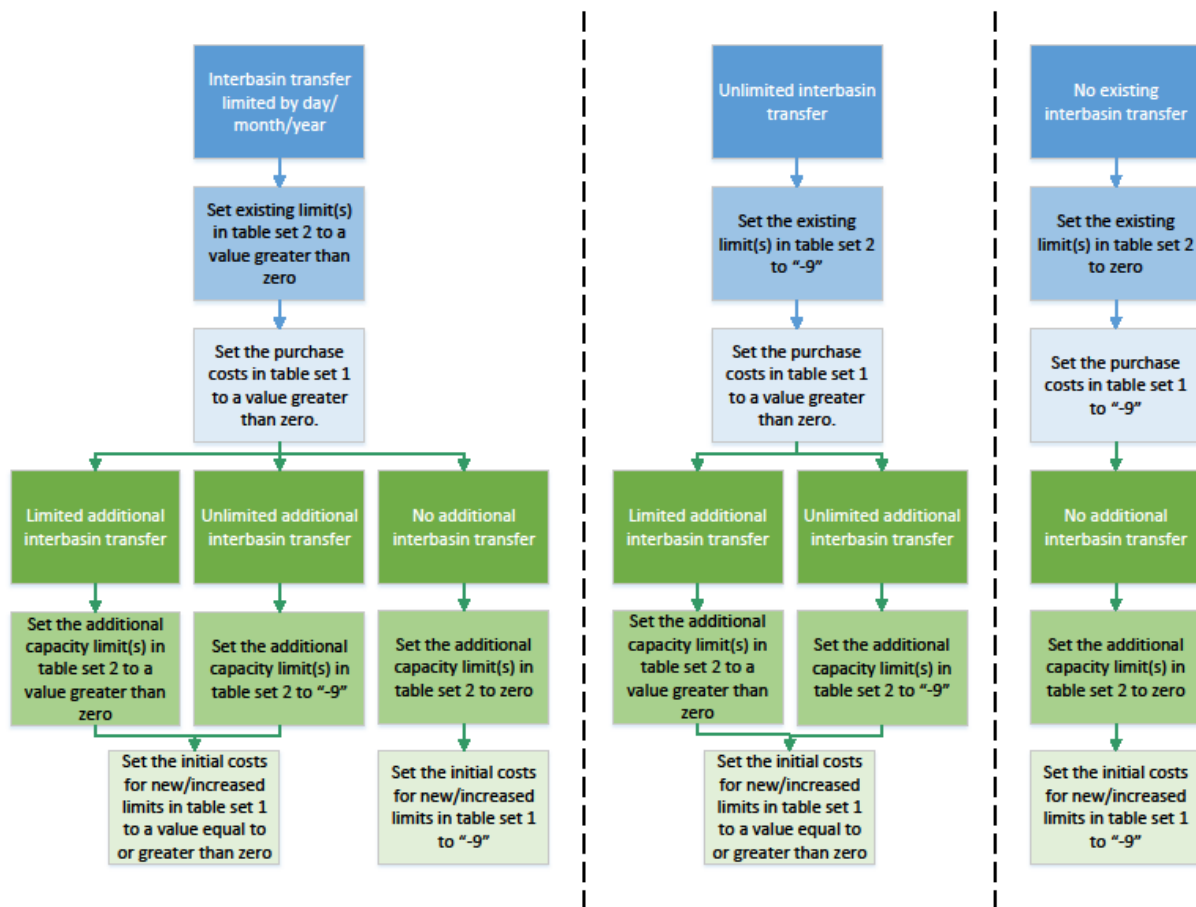
- and table set 2 related to capacity limits on interbasin transfer.

Enter existing limits on IBT for daily, monthly and/or annual basis. If flow is not limited, enter -9.			
Existing Limits on IBT [MG per month]			Existing Limits on IBT
Month	Water	Wastewater	
January	-9	-9	
February	-9	-9	
March	-9	-9	
April	-9	-9	
May	-9	-9	
June	-9	-9	
July	-9	-9	
August	-9	-9	
September	-9	-9	
October	-9	-9	
November	-9	-9	
December	-9	-9	

Additional Capacity Limits		
	Water	Wastewater
Daily [MGD]	3	1
Annual [MG per year]	-9	-9

If your watershed currently transfers water in or out, use the following diagram steps in blue to understand how to represent the *existing capacity* limitations and costs. If you would like to explore adding *additional capacity* to existing interbasin transfer agreements, use the following diagram steps in green to understand how to represent the *additional capacity* limitations and costs.

³ WMOST uses the concentration within the reservoir (XResF) to calculate loadings leaving the reservoir (LSwExt) so there should be no difference between the concentration within the reservoir and the concentration leaving the reservoir at the outlet of the watershed.



You should only set existing limit(s) at one temporal scale (*i.e.*, daily, monthly, or annually). All other limit(s) should be set to -9 as shown in the diagram.

You do not have to provide limits on IBT that match the time step of your model. For example, if you are running a daily model, you can provide limits on a monthly or annual basis. As noted in the user guide, the daily, monthly, and/or annual limits⁴ are adjusted accordingly.

⁴If you provide IBT limits expressed on multiple bases, make sure that the limits are not contradictory or incompatible because this would result in an infeasible model.